

Review

1. $5^{-2x} = 3$

$$\therefore \log_5 3 = 2x$$

$$x = \frac{\log_5 3}{2}$$

$$x = 0.341$$

2. $\cos \theta = \cos 295^\circ \Rightarrow \cos \theta = 65^\circ \therefore \theta = -65^\circ, 65^\circ$

3. $2x - 3y = 12 \Rightarrow y = \frac{12-2x}{3}$

Thus:

$$2x^2 - 3\left(\frac{12-2x}{3}\right) = 12$$

$$2x^2 + 2x - 24 = 0$$

$$(2x + 8)(x - 3) = 0$$

Therefore $x = 3$ or $x = -4$

When $x = 3, y = -2$

And when $x = -4, y = -\frac{20}{3}$

4. In $(a - b)^5$, the coefficients are 1, 5, 10, 10, 5, 1

So the coefficient of the term in a^2 is $10b^3$

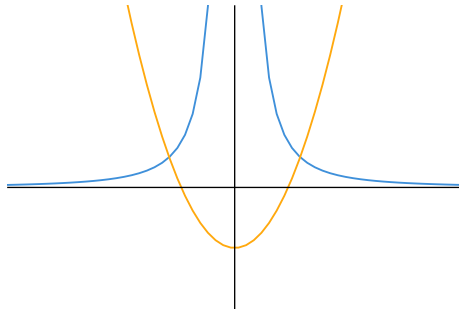
And the coefficient of the term b^4 is $5a$.

5. Graph S.

Exam Questions

1.

a.



b. $\therefore \frac{3}{x^2} = x^2 - 2$

$$3 = x^4 - 2x^2$$

$$0 = x^4 - 2x^2 - 3$$

$$0 = (x^2 - 3)(x^2 + 4)$$

$$\therefore x = \sqrt{3} \text{ or } x = -\sqrt{3}$$

$x^2 + 4$ does not factorize (the discriminant $b^2 - 4ac = 0 - 4 \times 1 \times 4 = -16$. $-16 < 0$)

$\therefore$ points of intersection are $(\sqrt{3}, 1)$ and $(-\sqrt{3}, 1)$.

2.

